

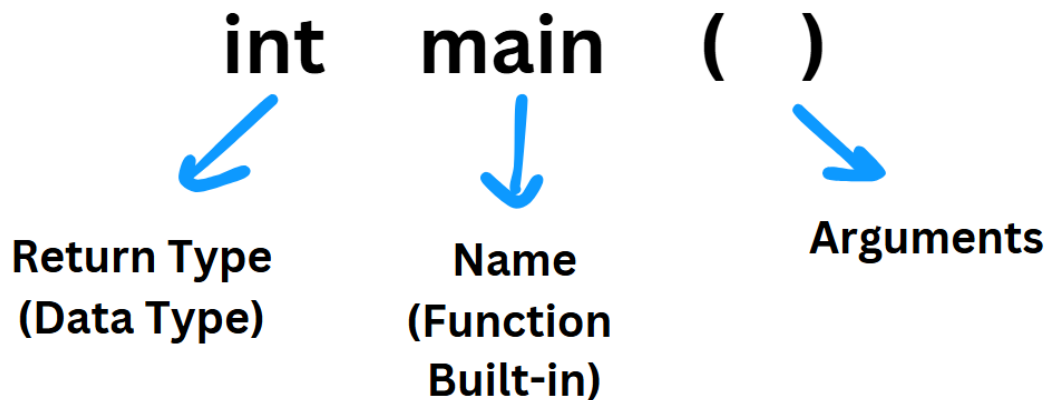
ITC:

ITC:

Begin:

Install Visual Studio for C++ (CTRL + F5) to Run Code.

- '#' are called pre-processor directives
- **Compiler** checks the syntax and converts (machine code → object program)
- **Linker** combines object program to create (.exe) file
- **IDE** has pre-written programs in library.
- **Loader** loads executable file into memory and executes to program.



- "{ " → Begin Body
- " cout " → console output
- "<< " → Stream insertion (into console)
- "\n " → escape sequence
- " endl " → get to next line (endline)
- "; " → statement terminator

ITC:

- "return 0"
- system("pause") → To hold exe File on screen.

Escape Sequences:

- "\n" → New Line
 - "\t" → move cursor 8 spaces next
 - "\a" → Alert sound the system beep
 - "\\" → Backslash used to print chracter
 - "\"" (Single Quote) and "\"" (Double Quote)
-

Keywords:

int, float, double, char, const, void, return (pre-defined compiler commands)

Identifiers:

consist of letters, digits and underscore chracter. must begin with a letter or underscore

(variables, constants and functions) e.g 'a1' or '_a'  but not '1a' 

variable = 2 → Constant (Number)

variable = 'A' → Single Chracter

variable = "Apple123" → Collection of chracters (String)

Operators:

Arithmetic:	Relational	Logical

ITC:

+, -, *, / , %	>, <, >=, <=, ==, !=	&&, , !
-------------------	----------------------	---------------

← Left to Right Precedence in C++

High: / * %

Low: + -

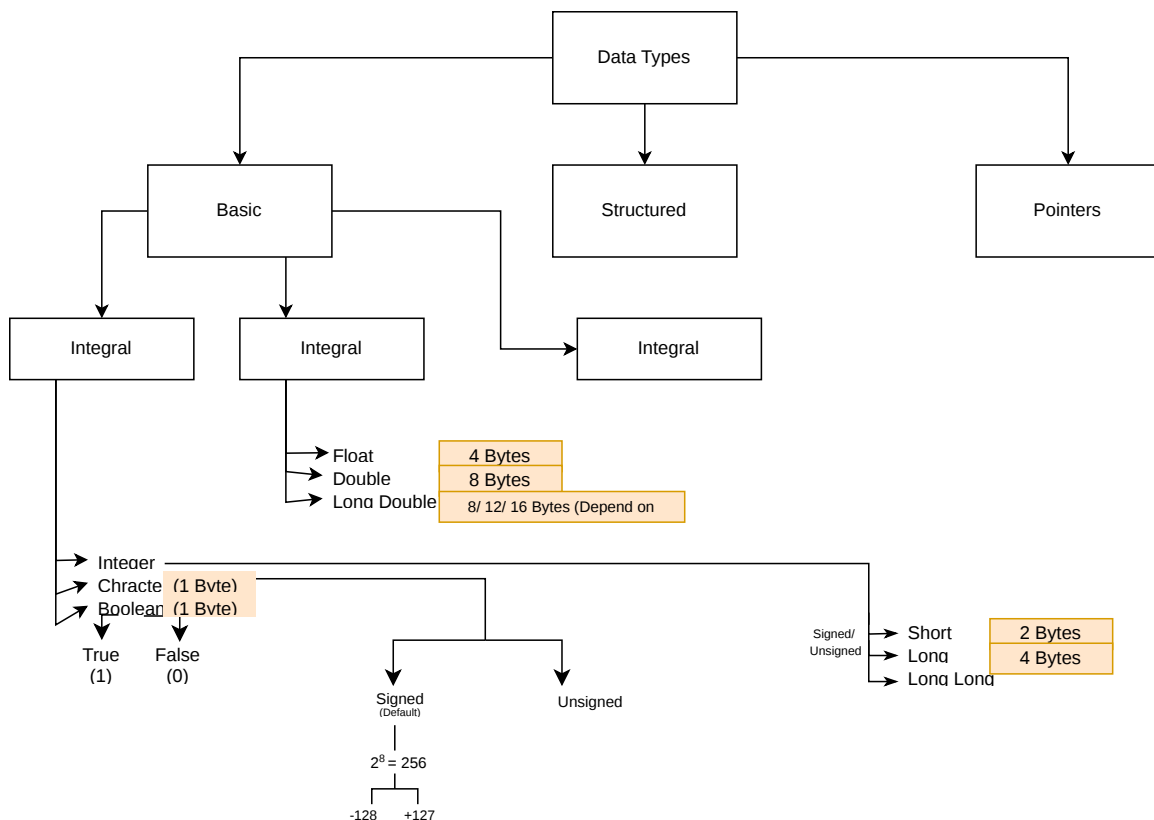
Example: `cout << 3 - 4 + 2 / 3 % 2;`

$3 - 4 + 1$

$-1 + 1 \rightarrow = 0$

AND OR XOR
NOT

F	F	F
F	T	F
T	F	F
T	T	T



ITC:

Data Types → MAX Values:

```
cout << "Minimum value of short integer: " << SHRT_MIN << endl;
cout << "Maximum value of short integer: " << SHRT_MAX << endl;
cout << "Maximum value of unsigned short integer: " << USHRT_MAX << endl;
cout << "Minimum value of integer: " << INT_MIN << endl;
cout << "Maximum value of integer: " << INT_MAX << endl;
cout << "Minimum value of unsigned integer: " << UINT_MAX << endl;
cout << "Minimum value of long integer: " << LONG_MIN << endl;
cout << "Maximum value of long integer: " << LONG_MAX << endl;
cout << "Maximum value of unsigned long integer: " << ULONG_MAX << endl;
cout << "Minimum value of long long integer: " << LLONG_MIN << endl;
cout << "Maximum value of long long integer: " << LLONG_MAX << endl;
cout << "Maximum value of unsigned long long integer: " << ULLONG_MAX << endl;
```

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.....

[LAB TASKS]

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ITC:

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TASK 1:

```
/******  
Write a Program that takes input your salary and output the per month
```

ITC:

```
profit,
if no amount is drawn from your account for 6 months

*****

#include <iostream>
using namespace std;

int main()
{
    float salary;

    cout << "Input Monthly Salary : ";
    cin>> salary;

    float profit;
    profit = salary * (0.10 * 0.5);

    cout<<"if you keep this amount in account for 6 months, your profit at
10% will be: "<<profit<<endl;

    return 0;
}

/*
Formula: "Interest = p x r x t" where
p = Principle Amount,
r = Annual Interst Rate,
t = Months expressed in years 6/12 = 0.5
*/
```

TASK 2:

```
/******
Centigrade to Farenheight (Convert)
*****

#include <iostream>
using namespace std;

int main()
{
    float tempr;
```

ITC:

```
cout << "Enter Temperature in Centigrade : ";
cin >> tempr;

cout << "= " << (tempr * 9/5) + 32 << " Farenheight."<<endl;
system ("pause");

return 0;
}
```

TASK 3 :

```
/******
Number greater than 9999 and less than 100000. Sum of Digits
******/

#include <iostream>
using namespace std;

int main()
{
    int num;
    cout<<"Enter Number greater than 9999 and less than 100000 : ";
    cin>> num;

    if (num > 9999 and num < 100000){
        int a1;
        int a2;
        int a3;
        int a4;
        int a5;
        int a6;
        int sum = 0;

        a1 = num % 10;
        sum = sum + a1;
        num = num/10;

        a2 = num % 10;
        sum = sum + a2;
        num = num/10;

        a3 = num % 10;
        sum = sum + a3;
        num = num/10;
```

ITC:

```
a4 = num % 10;
    sum = sum + a4;
    num = num/10;

    a5 = num % 10;
    sum = sum + a5;
    num = num/10;

    a6 = num % 10;
    sum = sum + a6;
    num = num/10;

    cout << "Sum of Digits are : "<<sum<<endl;
}
else{
    cout <<"Invalid Number!"<<endl;
}

return 0;
}
```

TASK 4:

```
/*.....
Multiple IFs Statements
.....*/

#include <iostream>
using namespace std;

int main()
{
    int num;
    cout << "Enter Number : ";
    cin >> num;

    if (num >=5 && num <=10)
        cout<<"Correct.";

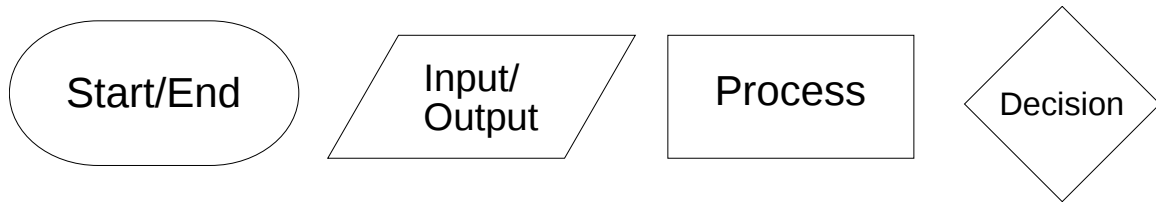
    if (num == 8)
        cout<<"Very Good.";

    if (num < 5)
        cout<<"Bad";
}
```

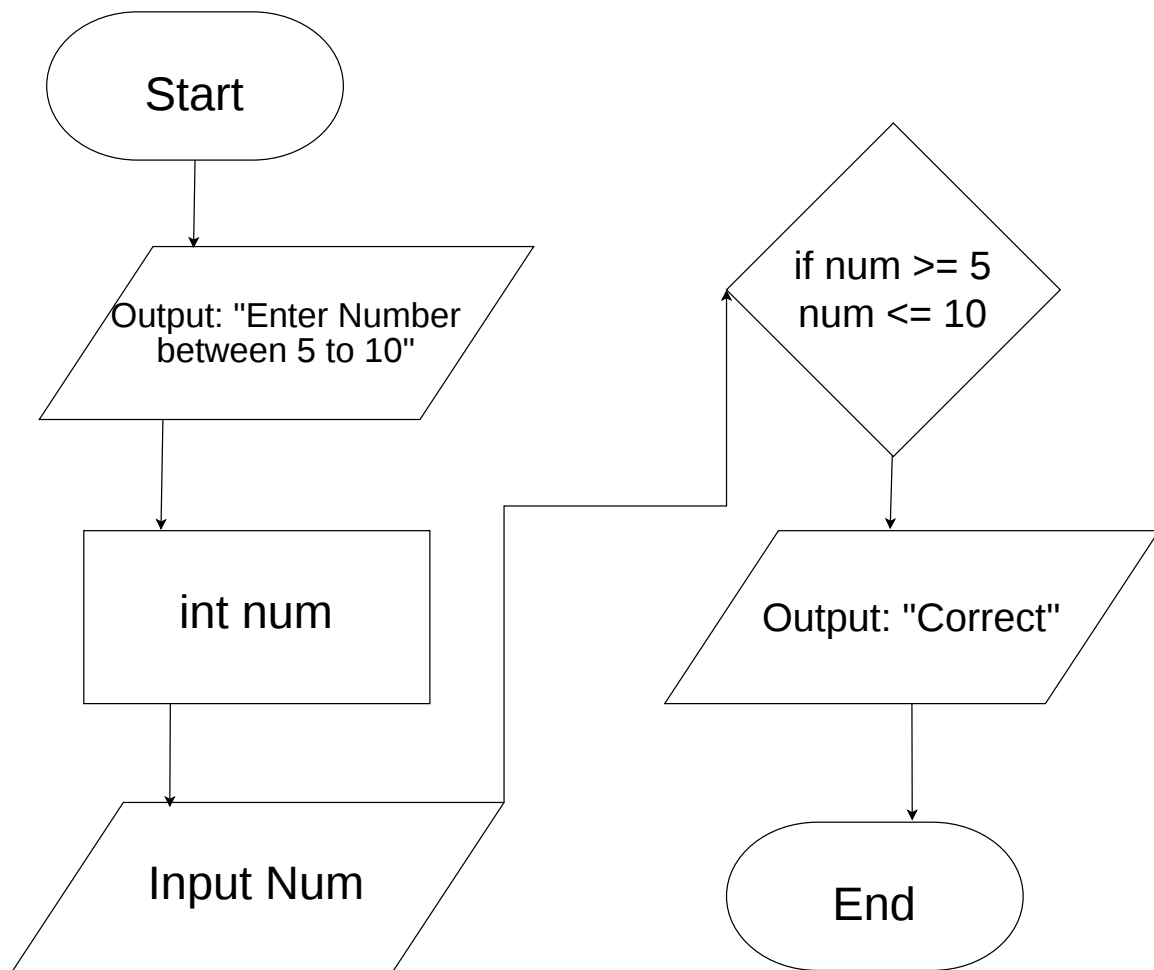

ITC:

```
return 0;  
}
```

FLOWCHART:



Example:



Ternary Operator:

if...else

```
#include <stdio.h>

int main() {
    int number = 3;

    if (number % 2 == 0) {
        printf("Even Number");
    }
    else {
        printf("Odd Number");
    }

    return 0;
}
```

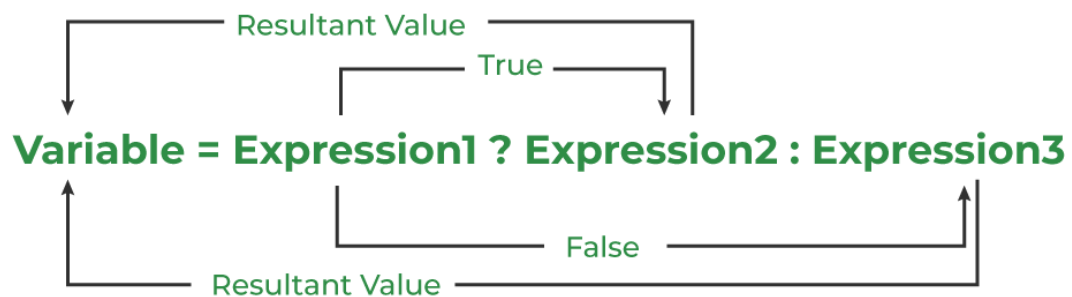
Ternary Operator

```
#include <stdio.h>

int main() {
    int number = 3;

    (number % 2 == 0) ?
        printf("Even Number") :
        printf("Odd Number");

    return 0;
}
```



```
Result = ( a > b ? " a > b " : " b > or = a" );
```

ITC:

TASK 5:

```

/*****
SWITCH Statement:
*****/

#include <iostream>
using namespace std;

int main()
{
    cout << "Enter A or B : ";
    char a;
    cin >> a;

    switch (a)
    {
        case 'A':
            cout << "You entered A\n";
            break;

        case 'B':
            cout << "You entered B\n";
            break;

        default:
            cout << " You Entered Non-Desired Character\n";

    }
    return 0;
}

/*
SYNTAX:
-----
switch (expression)
{
    case value1:
        statement1:
        break;

    case value2:
        statement2:
        break;
}

```

ITC:

```
...

    default:
        statement n;

}
-----
*/
```

TASK 6:

```
/******
Partial Initialization of Array:
******/

#include <iostream>
using namespace std;
#define S 3

int main()
{
    int i=0, a[S] = {23,43,55};

    while (i < 5){
        cout << a[i] << "\t";
        i++;
    }

    return 0;
}
```

TASK 7:

```
/******
Partial Initialization of Array:
******/

// (Reverse Traversal)

#include <iostream>
using namespace std;
#define S 3
```

ITC:

```
int main()
{
    int i = S-1;
    int a[S] = {23,43,55};

    while (i >= 0){
        cout << a[i] << "\t";
        i--;
    }

    return 0;
}
```

TASK 8

```
/*.....
Searching in Array (BUG)
*****
// <bits/stdc++.h>
#include <iostream>
using namespace std;

#define min 30
#define size 10

int main()
{
    int i=0, x, count=0, inp, arr[size]={23,45,55};

    while (i<size)
    {
        x = rand() % 100;
        cout << arr[i] << "\t";
        i++;

        cout <<"Enter Number between 0 to 100 : ";
        cin >> inp;

        for (int i = 0; i < size; i++) {
            cout << "\n" << arr[i] << " is item no. "<< i+1;
            count ++;
        }
    }
}
```

ITC:

```
if (count = 0)
{
    cout <<"\n Entered no. NOT FOUND !\n";
}

return 0;
}
```

TASK 9:

```
/******
Bubble SORT
*****

#include <iostream>
#include <ctime> // Use <ctime> instead of <time.h>
using namespace std;

#define SIZE 5

int main() {
    int i, j, temp;
    int arr[SIZE]; // Initialize the array without predefined values
    srand(time(0));

    // Fill the array with random numbers
    for (i = 0; i < SIZE; ++i) {
        arr[i] = rand() % 100; // Generate random numbers between 0 and 99
    }

    // Display the original array
    cout << "Original Array: \n";
    for (i = 0; i < SIZE; ++i) {
        cout << arr[i] << "\t";
    }
    cout << endl;

    // Bubble Sort Algorithm
    for (i = 0; i < SIZE - 1; i++) { // Loop through the array
        for (j = 0; j < SIZE - 1 - i; j++) { // Last i elements are already
sorted
            if (arr[j] > arr[j + 1]) {
```

ITC:

```
temp = arr[j];
        arr[j] = arr[j + 1];
        arr[j + 1] = temp;
    }
}

// Display the sorted array
cout << "After Bubble Sort: \n";
for (i = 0; i < SIZE; ++i) {
    cout << arr[i] << "\t";
}
cout << endl;

return 0;
}
```

TASK 10:

```
/*.....
Binary Search
.....*/

#include <iostream>
using namespace std;

int main() {
    int inp, arr[] = {23, 32, 44, 49, 55};
    int size = sizeof(arr) / sizeof(arr[0]); // Calculate the size of the
    array

    cout << "Enter Element to Search: ";
    cin >> inp;

    int first = 0, last = size - 1, mid;

    while (first <= last) {
        mid = (first + last) / 2; // Calculate mid at the beginning of the
        loop

        if (arr[mid] < inp) {
            first = mid + 1;
        }
    }
}
```

ITC:

```
cout << "Item found at position " << mid + 1 << endl;
    return 0; // Exit the program after finding the item
}
else {
    last = mid - 1;
}
}

// If the element is not found
cout << "Not Found!" << endl;

return 0;
}
```

TASK 11:

```
/******
Deletion (Array-Element)
******/

#include <iostream>
using namespace std;

int main() {

    int item, total=5, arr[5] = {23,34,33,44,55};
    //Show Array
    cout<<"Array Items : ";
    for (int i = 0; i < total; i++) {
        cout<<arr[i]<<" ";
    }
    cout<<endl;

    //Deletion Process
    cout<<"Delete: "; cin>>item;
    for (int i = 0; i < total; i++) {
        if(arr[i]==item){
            for(int j=i; j<(total-1); j++){
                arr[j] = arr[j+1];
            }
            total --;
        }
    }
}
```


ITC:

```
cout<<endl<<"Successfully Deleted! \nArray Items: ";
    for (int i = 0; i < total; i++) {
        cout<<arr[i]<<" ";
    }

    return 0;
}
```

TASK: 12

```
/*.....
Nested IF-Switch (Selection)
.....*/

#include<iostream>
using namespace std;
int main()
{
    cout << "Enter 1 for calculator \n Enter 2 to check positive/negative or
zero \t:";
    int selection;
    cin >> selection;

    if (selection == 2)
    {
        int input;
        cout<< "Enter a number to check for +ve, -ve or zero: ";
        cin >> input;
        if (input == 0) { cout << "You entered 0 \n"; }
        if (input < 0) { cout << "You entered negative \n"; }
        if (input > 0) { cout << "You entered positive \n"; }
    }
    else
    if (selection == 1)
    {
        float a, b;
        cout << "Enter first number : ";
        cin >> a;
        cout << "Enter second number : ";
        cin >> b;
        cout << "Enter +, -, *, / or % : ";
        int x = static_cast<int> (a);
        int y = static_cast<int> (b);
    }
}
```

ITC:

```
cin >> symbol;
switch (symbol)
{
case '+':
    cout << a << " + " << b << " = " << a + b << endl;
    break;
case '-':
    cout << a << " - " << b << " = " << a - b << endl;
    break;
case '*':
    cout << a << " * " << b << " = " << a * b << endl;
    break;
case '/':
    if (b != 0)
    {
        cout << a << " / " << b << " = " << a / b << endl;
        break;
    }
    else
    {
        cout << "The second number entered is 0. \n";
    }
case '%':
    cout << a << " % " << b << " = " << x % y << endl;
    break;
default:
    cout << "You didnt enter desired operation. \n";
}

}
system("pause");
return 1;
}
```

TASK 13:

```
/******
EvenOdd Count + Deletion
*****

#include<iostream>
using namespace std;
int main()
```

ITC:

```
srand (time(0));
int x,odd=0,even=0,total=25,count=0;
int arr[25] = {1,2,3,4};;

for (int i = 0; i < 25; i++) {
    x=rand() %51;
    if(x>10)
    {
        arr[i]=x;
    }

    if(arr[i] % 2 == 0){
        even++;
    }
    else if(arr[i]%2 != 0){
        odd++;
    }
    else
        continue;
}

//Display array
cout<<"\n\n\t\t[ Array Items: ]"<<endl;
for (int i = 0; i < total; i++) {
    cout << arr[i] <<" ";
}

cout<<"\n\nEven Numbers:  "<<even<<endl;
cout<<"Odd Numbers:  "<<odd<<endl;

int inp;
cout<<"\nEnter Number to Delete:  ";
cin>>inp;

for (int i = 0; i < total; i++) {
    if (arr[i] == inp){
        for(int j=i; j<(total-1); j++){
            arr[j] = arr[j+1];
        }
        total--;
        count++;
    }
}
```

ITC:

```
if (count == 0){
    cout << "\nNumber not found in array!";
}
cout << "\n\n\t\t[ Updated List ]" << endl;

for (int i = 0; i < total; i++) {
    cout << arr[i] << ", ";
}

return 0;
}
```

LAB Codes:

TASK 1:

```
/******
Middle Number (3 integers Input)
*****

#include <iostream>
using namespace std;
int main(){

    int num1;
    cout << "Enter 1st Number: ";
    cin >> num1;

    int num2;
    cout << "Enter 2nd Number: ";
    cin >> num2;

    int num3;
    cout << "Enter 3rd Number: ";
    cin >> num3;
    cout << endl;
```

ITC:

```
cout << "Middle Number is: " << num1 << endl;
}
if (num2 >= num1 && num2 <= num3 || num2 <= num1 && num2 >= num3){
    cout << "Middle Number is: " << num2 << endl;
}
if (num3 >= num1 && num3 <= num2 || num3 <= num1 && num3 >= num2){
    cout << "Middle Number is: " << num3 << endl;
}

system("pause");
return 0;
}
```

Task 2:

```
/*
Find Perimeter & Area of a rectangle.
Get Length / Width (150m/100m)

Formula (Rectangle):
Perimeter = 2 * (length + width)
Area = length * width

2. Take the radius from the user and calculate the area of the circle
Area =  $\pi r^2$ 
*/

#include <iostream>
using namespace std;

int main(){
    // Area & Perimeter of Rectangle
    cout << "[Calculate Area & Perimeter of Rectangle by Given Length and Width] \n" << endl;
    int Length;
    int Width;
    cout << "Enter Length= ", cin >> Length;
    cout << "Enter Width = ", cin >> Width;

    cout << "-> Area of Rectangle = " << Length * Width << " meter square" << endl;
    cout << "-> Perimeter of Rectangle = " << 2 * (Length + Width) <<
```

ITC:

```
cout << "_____\\n\\n" << endl;

// Area of Circle by given Radius
cout << "[Calculate Area of Circle by User Radius]\\n" << endl;
int Radius;
cout << "Enter Radius = ", cin >> Radius;
cout << "-> Area of Circle = " << 3.142*Radius*Radius << endl;
return 0;
}
```

TASK 3:

```
/******
Pyramids (without Loop)
*****

#include <iostream>
using namespace std;

// My Info
int main(){

    // My Info
    cout << "[My Information]\\n" << endl;
    cout << "Hello! \\nThis is my first Program. \\nMy Name is Afaq Ul Islam.
\\nMy Registration # is: L1F21BSCS0951\\n\\n" << endl;

    // Multiples of 2
    cout << "[Multiples of 2]\\n" << endl;
    cout << "2 x 1 = " << 2 * 1 << endl;
    cout << "2 x 2 = " << 2 * 2 << endl;
    cout << "2 x 3 = " << 2 * 3 << endl;
    cout << "2 x 4 = " << 2 * 4 << "\\n\\n" << endl;

    // Half Pyramid
    cout << "[Half Pyramid]\\n" << endl;
    cout << "* \\n* * \\n* * * \\n* * * * \\n* * * * * \\n\\n" <<
endl;

    // Inverted Half Pyramid
    cout << "[Inverted Half Pyramid]\\n" << endl;
    cout << "* * * * * \\n* * * * \\n* * * \\n* * \\n* \\n\\n" <<
endl;
```

ITC:

```
// Hollow Inverted Half Pyramid
cout << "[Hollow Inverted Half Pyramid]\n" << endl;
cout << "* * * * * \n*      * \n*      * \n*      * \n*      * \n\n" <<
endl;

// Full Pyramid
cout << "[Full Pyramid]\n" << endl;
cout << "      * \n      * * \n      * * * \n      * * * * \n\n" << endl;

// Inverted Full Pyramid
cout << "[Inverted Full Pyramid]\n" << endl;
cout << "* * * * * \n * * * * \n  * * * \n   * * \n    * \n\n" << endl;

// Hollow Full Pyramid
cout << "[Hollow Full Pyramid]\n" << endl;
cout << "      * \n      * * \n      * * * \n      * * * * \n\n" << endl;

return 0;
}
```

TASK 4:

```
/******
Arithmetic Operations
*****
#include <iostream>
using namespace std;
int main(){

    int a;
    cout << "-----" << endl;
    cout << "Enter your 1st number = ";
    cin >> a;

    int b;
    cout << "Enter your 2nd number = ";
    cin >> b;
    cout << "-----" << endl;
```

ITC:

```
int difference = a - b;
int product = a * b;
int quotient = a / b;
int reminder = a % b;

cout << "Sum = " << sum << endl;
cout << "Difference = " << difference << endl;
cout << "Product = " << product << endl;
cout << "Quotient = " << quotient << endl;
cout << "Reminder = " << reminder << endl;
cout << "-----" << endl;
cout << "\n";

return 0;
}
```

TASK 5:

```
/******
Second Max (Integer)
*****

#include<iostream>
using namespace std;

int main(){

    int a,b,c;
    cout << "Enter First Integer: ";    cin >> a;
    cout << "Enter Second Integer: "; cin >> b;
    cout << "Enter Third Integer: "; cin >> c;

    if (a >= b && a <= c || a <= b && a >= c){
        cout << "\n    ----> Integer (a) is 2nd Maximum Number = " << a <<
endl;
    }
    else if (b >= a && b <= c || b <= a && b >= c){
        cout << "\n    ----> Integer (b) is 2nd Maximum Number = " << b <<
endl;
    }
    else
        cout << "\n    ----> Integer (c) is 2nd Maximum Number = " << c <<
endl<<endl<<endl;
```


ITC:

```
return 0;
}
```

TASK 6:

```
/******
GradesResult (5 Courses)
******/

#include<iostream>
using namespace std;
int main(){

    int a,b,c,d,e;

    cout << "Enter Marks of Functional English: "; cin >> a;
    cout << "Enter Marks of ITC: "; cin >> b;
    cout << "Enter Marks of ITC LAB: "; cin >> c;
    cout << "Enter Marks of Basic Electronics: "; cin >> d;
    cout << "Enter Marks of Logical Thinking: "; cin >> e;

    float sum = a + b + c + d + e;
    float percentage = sum / 500 * 100;
    cout << "\n-----\n[ Total Percentage ] = " << percentage << "%" <<
endl;

    // TypeCast
    int per = static_cast<int>(percentage);

    switch (per / 10)
    {
        case 9:
            cout << "Your grade is: A\n";
            break;

        case 8:
            cout << "Your grade is: B\n";
            break;

        case 7:
            cout << "Your grade is: C\n";
            break;
```

ITC:

```
    cout << "Your grade is: D\n";
        break;
    case 5:
        cout << "Your grade is: E\n";
        break;

    case 4:
        cout << "Your grade is: F\n";
        break;

    default:
        cout << "You're Failed!\nYour percentage is below the passing
marks." << endl;
    }
    return 0;
}
```

TASK: 7

```
// Countdown till 1 (from InputNumber)

#include <iostream>
using namespace std;
int main() {

    int num;
    cout << "Enter Number: ";    cin >> num;

    while (num >= 1) {
        cout << num-- << endl;
    }
    system("pause");
    return 0;
}
```

TASK 8:

```
// Guessing Game (Using Flag-Controlled While loop).

#include <iostream>
#include <cstdlib>
#include <ctime>
```

ITC:

```
using namespace std;
int main()
{
    int num, guess, attempt, remainAttempt = 2;
    attempt = 1;
    bool isGuessed;
    srand(time(0));
    num = rand() % 5;
    isGuessed = false;

    while (attempt <= 2 && (!isGuessed))
    {
        cout << "Guess Number between 0 to 5 (" << remainAttempt << " Attempts
Left) : ";
        cin >> guess;
        remainAttempt--;

        if (guess == num){
            cout << "\nCorrectly Guessed :D \nThe Number is: " << num <<
endl<<endl;
            isGuessed = true;
        }
        else if (attempt <= 2){
            cout << "Try Again!\n" << endl;
            attempt++;
        }
    }

    if (!isGuessed) {
        cout << "YOU LOSE! , The Number was: " << num << endl<<endl;
    }
    system("pause");
    return 0;
}
```

TASK 9:

```
// Sum of Digits (using Sentinel-Controlled :While loop).

#include <iostream>
using namespace std;
int main() {
```

ITC:

```
int sum = 0;

cout << "Enter a 5 Digit Number: " << endl; cin >> num;
cout << endl;

if (num < 100000 && num > 9999)
{
    cout << "Sum of: ";

    while (num > 0){
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;

        cout << digit << "+";
    }
    cout << "= " << sum << endl;
}
else
    cout << "\nInvalid Input. Enter 5 Digit Number!\n" << endl;

system("pause");
return 0;
}
```

TASK 10:

```
// Sum of Digits (DO-While Loop)

#include <iostream>
using namespace std;
int main() {

    int num, digit;
    int sum = 0;
    cout << "Enter a 5 Digit Number: " << endl;
    cin >> num;
    cout << endl;

    if (num < 100000 && num > 9999){
        cout << "Sum of: ";
```

ITC:

```
{
    digit = num % 10;
    sum = sum + digit;
    num = num / 10;

    cout << digit << "+";

    } while (num > 0);
    cout << "= " << sum << endl;
}
else
    cout << "\nInvalid Input. Enter 5 Digit Number!\n" << endl;

system("pause");
return 0;
}
```

TASK 11:

```
// Array SUM,Avg

#include <iostream>
using namespace std;

int main()
{
    // Display 'Array Initialized with 10 Values.'
    int count=0;
    int arr[10] = {12,24,36,48,52,65,85,14,32,26};

    cout <<"Array : ";
    for(int i = 0; i < 10; i++)
        cout<< arr[i]<<" ";

    cout<<endl<<endl;

    // Sum of Numbers
    int sum = 0;

    for(int i = 0; i < 10; i++){
        if(arr[i] % 2 == 0)
        {
```

ITC:

```
count++;
    }
}
cout<<"Sum of Numbers: "<<sum<<endl;

// Average of Numbers
float avg=0;

for(int i = 0; i < 10; i++){
    if(arr[i] % 2 == 0)
    {
        avg = (float)sum/count;
    }
}
cout<<"Average of Numbers: "<<avg<<endl;

return 0;
}
```

TASK 12:

```
/* --- Prime / Not-Prime
An even number is prime if it is 2.
An odd integer is prime if it is not divisible by
(any odd integer less than or equal to the square root)
*/

#include <iostream>
using namespace std;
int main()
{
    int num,d,check=1;

    cout << "Enter Any +ve Integer: "; cin >> num;
    if (num == 0 || num == 1)
        check = 0;
    else
    {
        for (d = 2; d < num ; d++)
            if(num % d == 0)
```

ITC:

```
}

    if (check == 1)
        cout << "Prime Number." << endl;

    else
        cout << "Non-Prime Number" << endl;

    return 0;
}
```

TASK: 13

```
/* Right Triangle or Not ?

if (LHS == RHS means RightTriangle 'TRUE')
LHS = hyp^2
RHS = base^2 + perp^2
*/

#include <iostream>
using namespace std;
int main()
{
    int hyp, base, perp;

    cout << "Enter Hypotenuse Of Triangle: "; cin >> hyp;
    cout << "Enter Base Of Triangle: "; cin >> base;
    cout << "Enter Perpendicular Of Triangle: "; cin >> perp;

    if ( hyp * hyp == (base * base) + (perp * perp) )
        cout << "\nThis is a right angle triangle." << endl;
    else
        cout << "\nThis is not a right angle triangle." << endl;
    return 0;
}
```

TASK 14:

```
// Integer to Binary && ASCII to Integer
#include <iostream>
using namespace std;
```

ITC:

```
int main()
{
    int select;
    cout << "Press 1 for [ ASCII --> INTEGER ]\n";
    cout << "Press 2 for [ INTEGER --> BINARY ]\n";
    cout << "Exit 0\n";
    cout << "\n Select: "; cin>>select;

    while(1)
    {
        if(select == 1)
        {
            char chr;
            cout << "[ ASCII --> INTEGER ]\n";
            cout << "_____ \n";
            cout << "Enter your Character: ";
            cin >> chr;
            cout << "Integer value of " << chr << " is: " << (int)chr <<
endl;

            cout << endl;
            break;
            /*****/
        }
        else if(select == 2)
        {
            cout << "\n\n[ INTEGER --> BINARY ]\n";
            int input,i;
            bool bin[8]={};

            cout<<"Enter Number (0 - 255) : ";
            cin>>input;

            for(i=0;i<8;i++) {
                bin[i]=input%2;
                input=input/2;}

            cout<<"Binary: ";
            for(i=0;i<8;i++){
                cout<<bin[i];
            }
            break;
        }
        else
            break;
    }
}
```


ITC:

```
}  
  
    return 0;  
}
```

TASK 15:

```
// Find Substring in a String.  
  
#include<iostream>  
using namespace std;  
#define SIZE 50  
  
int main()  
{  
    char str[SIZE], sbstr[SIZE / 2];  
  
    cout << "Enter a line to save: ";  
    cin.getline(str, SIZE);  
    cout << "Enter a sub-string to search in line: ";  
    cin.getline(sbstr, SIZE / 2);  
  
    int count = sizeof(sbstr), a = 0, sbstrLen=0, startindex=-1;  
  
    for (int i = 0; sbstr[i] != '\0'; i++)  
        sbstrLen++;  
  
    for (int i = 0; str[i] != '\0'; i++)  
    {  
        if (sbstr[0] == str[i])  
        {  
            startindex++;  
            a++;  
            for (int j = 1; sbstr[j] != '\0'; j++)  
            {  
                if (sbstr[j] == str[j + i])  
                {  
                    a++;  
                }  
            }  
        }  
    }  
}
```

ITC:

```
if (a== sbstrLen)
    cout << "\n"<<sbstr << " found from index "<<startindex<< " to index
"<<startindex+sbstrLen;

    return 0;
}
```

TASK 16:

```
// Generate Rand() Characters, Store in Char Array (Find SecondMinimum
Character)

#include<iostream>
#include<time.h>
#define SIZE 15
using namespace std;
int main()
{
    char arr1[SIZE], fmin, smin;
    int i = 0;
    srand(time(0));

    cout << "Random Generated Characters: \n";

    while (i < SIZE){
        arr1[i] = rand();
        if (arr1[i] >= 65 && arr1[i] <= 90)
        {
            cout << arr1[i]<< ", ";
            i++;
        }
        else
            continue;
    }
    fmin = smin = 'Z';
    cout << "\nSecond Minimum: ";
    for (i = 0; i < SIZE;i++)
    {
        if (arr1[i] < fmin)
        {
            smin = fmin;
            fmin = arr1[i];
        }
    }
}
```

ITC:

```
else
    if (arr1[i] < smin && arr1[i] != fmin)
        smin = arr1[i];
    }
    cout << smin<<endl<<endl;
    system("pause");
    return 0;
}
```

TASK 17:

```
// String Copying

#include<iostream>
using namespace std;
int main()
{
    char str1[100], str2[100];
    int i;

    cout << "Enter String: ";
    cin.getline(str1, 100);

    for (i = 0; str1[i] != '\0'; i++)
        str2[i] = str1[i];

    str2[i] = '\0';
    cout << "\nString 1: " << str1<<endl;
    cout << "String 2: " << str2 << " (Copied)"<<endl<<endl;

    system("pause");
    return 0;
}
```

TASK 18:

```
// Convert String characters (UpperCase to LowerCase and Vice versa)

#include <iostream>
using namespace std;

int main() {
```

ITC:

```
char str1[100];
int i;

cout << "Enter String: ";
cin.getline(str1, 100);

for (i = 0; str1[i] != '\0'; i++) {
    // Check uppercase
    if (str1[i] >= 'A' && str1[i] <= 'Z') {
        str1[i] = str1[i] + 32;
    }
    // Check lowercase
    else if (str1[i] >= 'a' && str1[i] <= 'z') {
        str1[i] = str1[i] - 32;
    }
}

cout << "Inverted String: " << str1 << endl;
return 0;
}
```

TASK 19:

```
// Count Vowels (String)

#include <iostream>
using namespace std;

int main() {
    char str[100];
    int count = 0;

    cout << "Enter a string: ";
    cin.getline(str, 100);

    for (int i = 0; str[i] != '\0'; i++) {
        if (str[i] == 'A' || str[i] == 'E' || str[i] == 'I' || str[i] == 'O'
        || str[i] == 'U' ||
            str[i] == 'a' || str[i] == 'e' || str[i] == 'i' || str[i] == 'o'
        || str[i] == 'u') {
            count++;
        }
    }
}
```

ITC:

```
return 0;
}
```

IMPORTANT:

TASK 20:

```
// Most Occuring Number
#include <iostream>
using namespace std;

int main()
{
    int x, arr[5] = {1,2,1,3,4};
    cout<< "Most Occuring Item: ";

    for (int i = 0; i < 5; i++) {
        for(int j=1; j<5; j++){
            if(arr[i] == arr[i+j])
                x = arr[i];
        }
    }

    cout <<x<<endl;
    return 0;
}
```

TASK 21:

```
/*
1
2 4
3 6 9
4 8 12 16
5 10 15 20 25
*/

#include <iostream>
using namespace std;

int main()
{
    for (int i = 1; i <=5 ; i++) {
```

ITC:

```
for(int j=1; j<=i; j++){
    cout<< i*j<< " ";
}
cout<<endl;
}
return 0;
}
```

TASK 22:

```
// SORTING

#include <iostream>
using namespace std;

int main()
{
    int num, SIZE=5;
    int arr[SIZE] = {4,1,5,3,2};

    // SORTING
    for (int i=0 ; i<SIZE; i++) {
        for (int j=i+1 ; j<SIZE; j++) {
            if (arr[j]<arr[i]){
                int temp = arr[i];
                arr[i]=arr[j];
                arr[j]=temp; }
        }
        cout<<arr[i]<<" ";
    }
    return 0;
}
```

TASK 23:

```
// Linear Search

#include <iostream>
using namespace std;

int main()
{
```

ITC:

```
int inp, arr[5] = {1,2,3,4,5};

    for (int i = 0; i < 5; i++) {
        cout << arr[i]<< " ";
    }
    cout<<"\nEnter Number to Find: "; cin>>inp;

    for (int i = 0; i < 5; i++) {
        if(arr[i]==inp)
            cout<<"Item: "<<i+1<<" at index "<<i<<endl;
    }
    return 0;
}
```

TASK 24:

```
// Insert Number in Array

#include <iostream>
using namespace std;

int main()
{
    int i, arr[6], elem;
    cout<<"Enter 5 Array Elements: ";
    for (i = 0; i < 5; i++)
        cin>>arr[i];

    cout<<"Insert Element: ";
    cin>>elem;

    arr[i] = elem;
    cout<<"\n---- Updated Array: ---- \n";

    for (i = 0; i < 6; i++)
        cout<<arr[i]<<" ";

    cout<<endl;
    return 0;
}
```

ITC:

TASK 25:

```
/*
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
*/

#include <iostream>
using namespace std;

int main()
{
    int i,j,a;
    for (i = 0; i < 5; i++) {
        a=1;
        for(j=0; j<=i; j++){
            cout<<a<<" ";
            a++;
        }
        cout<<endl;
    }
    cout<<endl;
    return 0;
}
```

TASK 26:

```
/*
*
* *
* * *
* * * *
* * * * *
*/

#include <iostream>
using namespace std;

int main()
{
```


ITC:

```
int i,j;
for (i = 0; i <= 5; i++) {
    for(j=0; j<i; j++){
        cout<<"* ";
    }
    cout<<endl;
}

return 0;
}
```

TASK 27:

```
// Sort Even then Odds (in Array)

#include <iostream>
using namespace std;

int main()
{
    int even, arr[10]= {1,2,3,4,5,6,7,8,9,10};

    for (int i = 0; i < 10; i++) {
        for(int j=0; j<10; j++){
            if(arr[i]%2 == 0){
                even = arr[i];
                arr[i]=arr[j];
                arr[j]=even;
            }
        }
    }

    for(int i=0; i<10; i++)
        cout<<arr[i]<<" ";

    return 0;
}
```

TASK 28:

```
// Putting 0s and 1s first in an Array.
```

ITC:

```
#include <iostream>
using namespace std;

int main()
{
    int temp, arr[10] = {1,2,1,0,5,6,7,1,9,10};

    for (int i = 0; i < 10; i++) {
        for(int j=0; j<10; j++){
            if(arr[i] == 0 || arr[i] == 1){
                temp = arr[i];
                arr[i]= arr[j];
                arr[j]= temp;
            }
        }
    }

    for (int i = 0; i < 10; i++)
        cout<<arr[i]<<" ";

    return 0;
}
```

TASK 29:

```
// Replace (Number)Item in Array.

#include <iostream>
using namespace std;

int main()
{
    int input, size = 5, arrOriginalSize=5;
    int arr[size] = {1,2,3,4,5};

    cout<<"Original Array: "<<endl;
    for (int i = 0; i < size; i++) {
        cout<<arr[i]<<" ";
    }

    int select;
    cout<<"\n\nSelect Item: "; cin>>select;
```

ITC:

```
if(arr[i] == select){
    cout<<"Replace by: "; cin>>input;
    arr[i] = input;
}
}

//Updated Array:
cout<<"\n_____ \nUpdated Array: "<<endl;
for (int i = 0; i < size; i++) {
    cout<<arr[i]<<" ";
}

return 0;
}
```

TASK 30:

```
// Test Scores of 10 Students, Find Avg, Output Scores (less than Avg)

#include<iostream>
using namespace std;
int main()
{
    float a[10],sum = 0,percentage;

    cout<<"Enter Marks of 10 Students: ";

    for(int i=0; i<10; i++){
        cin>>a[i];
        sum = sum + a[i];
    }
    percentage = sum / 10;
    cout<<"\nClass Average Percentage: "<<percentage<<"%"<<endl;

    // less than average.
    cout<<"Marks Less than Average: [";
    for(int i=0; i<10; i++){
        if(a[i]<= percentage)
            cout<<a[i]<<" ";
    }
    cout<<"]";
}
```

ITC:

```
return 0;
}
```

TASK 31:

```
// CGPA Calculator

#include <iostream>
using namespace std;
int main()
{
    cout << "\t YOU HAVE ONLY FOLLOWING GRADES\n";
    cout << "\t( A  A-  B+  B  C+  C  D+  D  E  )\n";
    int A = 4.0;
    int A_minus = 3.6;
    int B_plus = 3.3;
    int B = 3.0;
    int C_plus = 2.5;
    int C = 2.0;
    int D_plus = 1.5;
    int D = 1.0;
    int E = 0.0;
    string grade;
    float credit, total_credit = 0, time, total_time = 0, option, final_GPA;
    for (;;)
    {
        cout << "\nSubject:\nEnter Grade : "; cin >> grade;
        cout << "Credit Hours: "; cin >> credit;
        if (grade == "a" || grade == "A")
            time = credit*A;
        else if (grade == "a-" || grade == "A-")
            time = credit*A_minus;
        else if (grade == "b+" || grade == "B+")
            time = credit*B_plus;
        else if (grade == "b" || grade == "B")
            time = credit*B;
        else if (grade == "c+" || grade == "C+")
            time = credit*C_plus;
        else if (grade == "c" || grade == "C")
            time = credit*C;
        else if (grade == "d+" || grade == "D+")
            time = credit*D_plus;
        else if (grade == "d" || grade == "D")
            time = credit*D;
    }
}
```

ITC:

```
time = credit*D;
    else if (grade == "e" || grade == "E")
        time = credit*E;
    else
        cout << "INVALID CHOICE... TRY AGAIN!\n";
        total_credit = total_credit + credit;
        total_time = total_time + time;
        cout << "\nDo you want to enter another grade of subject (1 for YES
, 2 for NO) : "; cin >> option;
        if (option == 1)
        {
            continue;
        }
        else
        {
            break;
        }
    }
    final_GPA = total_time/total_credit;
    cout << "\t ----> Student's GPA is : " << final_GPA << endl;
    return 0;
}
```

TASK 32:

```
// String Array vs ChracterArray (Reverse)
#include<iostream>
using namespace std;

int main()
{
    char arr1[ ]={"HelloWorld"},arr2[ ]={'H','e','l','l','o','\0'};
    int i,size1=sizeof(arr1),size2=sizeof(arr2);

    cout<<"arr1 size is: "<<size1<<"\t";
    cout<<"arr2 size is: "<<size2<<endl;

    // Array 1_____
    cout<<"\nCharacter Array arr1 with loop is: ";
    for(i=0;i<size1;i++)
    {cout<<arr1[i];}

    cout<<"\nCharacter Array arr1 without loop is: ";
    cout<<arr1;
```

ITC:

```
cout<<"\nCharacter Array arr1 with loop in reverse is: ";
    for(i=(size1-1);i>=0;--i)
        cout<<arr1[i];

    // Array 2_____
    cout<<"\n\nCharacter Array arr2 with loop is: ";
    for(i=0;i<size2;++i)
        cout<<arr2[i];

    cout<<"\nCharcter Array arr2 without loop is: ";
    cout<<arr2;

    cout<<"\nThe Character Array arr2 with loop in reverse is: ";
    for(i=(size2-1);i>=0;--i)
        cout<<arr2[i];

    cout<<endl;
    return 0;
}
```

TASK 33:

```
// Input Continue, until Odd Number is Entered!

#include<iostream>
using namespace std;
int main()
{
    int input = 0;
    bool check = false;

    while(check != true) {
        cout<<"\n\nEnter a number: "; cin>> input;

        if(input % 2 == 0){
            cout<<"Even Number";
            continue;
        }
        if(input % 2 != 0){
            cout<<"Odd Number";
            break;
        }
    }
}
```

ITC:

```
return 0;
}
```

TASK 34:

```
/*
*      * * * * *
*      *
*      *
*      *
* * * * * * * * *
*          *      *
*          *      *
*          *      *
* * * * *      *
*/
#include<iostream>
#include<string>
using namespace std;
int main()
{
    cout << "      * * * * *\n";
    string x = "      *";
    string z = "      *      *";
    int y=1;
    while (y<4)
    {
        cout << x << endl;
        y++;
        if (y == 4)
            cout << " * * * * * * * * \n";
        while (y>3&& y<7)
        {
            cout << z << endl;
            y++;
        }
    }
    cout << " * * * * *      *\n";
    return 0;
}
```

TASK 35:

ITC:

```
// Numinator/Denominator (Breakdown)

#include<iostream>
using namespace std;
int main()
{
    int Numinator = 1;
    int Denomninator = 2;
    int nthTermNuminator, nthTermDenominator;

    cout<<"Enter nth term (Numinator & Denomninator): ";
    cin >> nthTermNuminator >> nthTermDenominator;

    int i=1, j=2;

    while(i <= nthTermNuminator && j<= nthTermDenominator)
    {
        cout<<i<<"/"<<j<<" + ";
        if( i==nthTermNuminator && j==nthTermDenominator ) cout<<"..."
<<endl;
        i++;
        j+=2;
    }
    return 0;
}
```

TASK 36:

```
// 5-Digit Number, Reverse it, Check if Input is Less than Reverse
#include<iostream>
using namespace std;
int main()
{
    int x,inp, rinp = 0;
    int i, temp,b = 10;

    cout<<"Enter a 5 digit number: ";
    cin>>inp;
    temp = inp;    // store input for compare.
```


ITC:

```
{

    cout<<"Reverse of "<<inp<<" is: ";
    for(i=0; i<5; i++)
    {
        rinp = rinp + (inp % b);
        rinp *= 10;
        inp = inp/b;
    }
    rinp /= 10;
    cout<<rinp<<endl;

    if(rinp > temp)
        cout<<"\nInput is Less than Output.\n";
    else if (rinp == temp)
        cout<<"\nInput is Equal to Output.\n";
    else
        cout<<"\nInput is Greater than Output.\n";
}
else
    cout << "\n\nWrong Input!\n\n"<<endl;
return 0;
}
```

TASK 37:

```
// (20% Discount) on above 1000$ AND Buy more than 20 Products
// (10% Discount) on above 1000$ OR Buy more than 20 Products
#include <iostream>
using namespace std;
int main()
{
    int qty, price, totalprice;
    cout << "Enter the Quantity of Product :"; cin >> qty;
    cout << "Enter the Price of Product :"; cin >> price;

    if (qty >= 20 && price >= 1000) {
        totalprice = qty * price;
        cout << "\n(20% Discount) Total = " << totalprice * 0.2 << endl;
        cout << endl;
    }
    else if (qty >= 20 || price >= 1000) {
```

ITC:

```
cout << "\n(10% Discount) Total = " << totalprice * 0.1 << endl;
cout << endl;
}
else {
    totalprice = qty * price;
    cout << "\n(0% Discount) Total Price = " << totalprice << endl;
    cout << endl;
}
return 0;
}
```

TASK 38:

```
// Postfix, Prefix VariableLogics
#include <iostream>
using namespace std;

int main()
{
    int x, w;
    char y, v;
    float z;

    x = 3.9/4-8+6.2*1.5;
    cout <<"x = "<<x<<endl;

    y = 66 + x++ + ++x;
    cout <<"y = "<<y<<endl;

    z = y++ * 0.5;
    cout <<"z = "<<z<<endl;

    w = (static_cast<int> (z)%11);
    cout<<"w = "<<w<<endl;

    v = (static_cast<int> (z)*3-8);
    cout<<"v = "<<v<<endl;

    return 0;
}
```

TASK 39:

ITC:

```
// Table Generator
#include <iostream>
using namespace std;
int main()
{
    int n, i = 1;
    cout << " Print table of any number from 2 to 20 " << endl;
    cout << " _____ \n" << endl;
    cout << " Enter a Number: ";
    cin >> n;

    if ((n > 1 && n <= 20))
    {
        while(i<=10){
            cout << endl;
            cout << n << " x " << i << " = " << n*i;
            i++;
        }
    }
    else{
        cout << "Invalid Number, Enter Number from (2 to 20)";
        return 0;
    }
}
```